Дати за изпити: 28.06. (пн) – Писмен изпит
02.07. (пт) – Теория (сутрин)

07.05 (петък) от 10:30 – Второ контролно

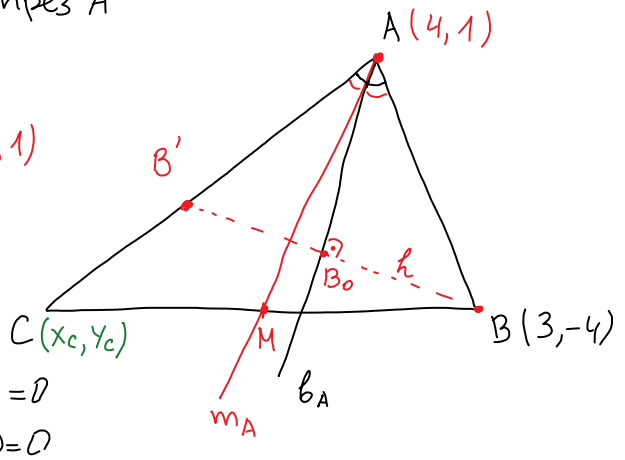
Зад. ОКС Oxy

$b_A: 2x-3y-5=0$

$m_A: x-8y+4=0$

т. $B(3,-4)$

а) ? коорд. на върховете A и C на $\triangle ABC$, за който
b_A – вътр. ъглополяр. при т. A
m_A – медиана през A



Решение:

1) $т.A = b_A \cap m_A \Rightarrow \begin{vmatrix} 2x-3y-5=0 \\ x-8y+4=0 \end{vmatrix} \Rightarrow A(4,1)$

2) Ако т. $B \xrightarrow{b_A} B'$, то $B' \in AC$

$h \begin{cases} \in B(3,-4) \\ \perp b_A: 2x-3y-5=0 \\ \quad\; 3x+2y \end{cases} \Rightarrow h: 3x+2y+D=0$

$3.3+2.(-4)+D=0$

$D=-1$

$h: 3x+2y-1=0$

$B_0 = h \cap b_A \Rightarrow \begin{vmatrix} 3x+2y-1=0 \;/.3 \\ 2x-3y-5=0 \;/.2 \end{vmatrix} + \Rightarrow 13.x-13=0 \Rightarrow x=1$

$y=-1$

$B_0(1,-1)$ – средата

$B(3,-4)$
$B'(x',y')$ $\Rightarrow \frac{x'+3}{2}=1 \qquad \frac{y'+(-4)}{2}=-1$

$B'(-1,2)$

3) $AC \equiv AB' \begin{cases} \in A(4,1)! \\ \in B'(-1,2)! \end{cases} \Rightarrow \begin{vmatrix} x & y & 1 \\ 4 & 1 & 1 \\ -1 & 2 & 1 \end{vmatrix} = 0 \qquad -x-5y+9=0 \;/.(-1)$

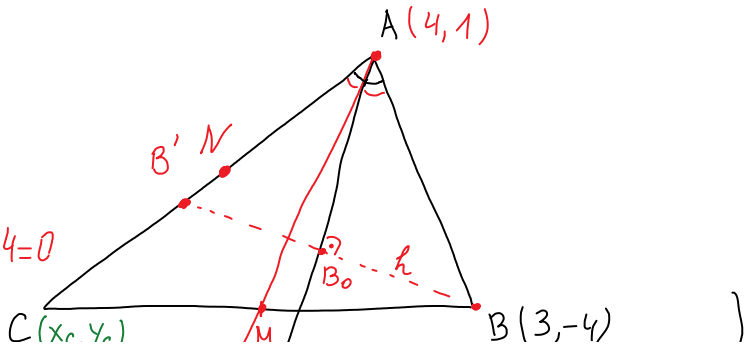
$AC: x+5y-9=0$ – ОА

4 1
−1 2

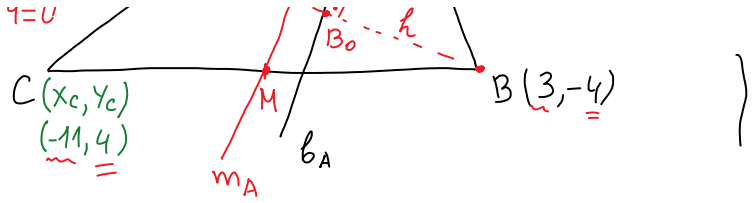
4) $C(x_c, y_c)$

$C \in AC \Rightarrow \boxed{x_c+5y_c-9=0}$ (1)

$M\left(\frac{x_c+3}{2}, \frac{y_c+(-4)}{2}\right) \rightarrow m_A: x-8y+4=0$



$\frac{x_C+3}{2} - 8\cdot\frac{(y_C-4)}{2} + 4 = 0$ (2) (2)



$x_C + 3 - 8y_C + 12 + 8 = 0$

$x_C - 8y_C + 43 = 0$ (2)

$\Rightarrow C: \begin{vmatrix} x_C + 5y_C - 9 = 0 \\ x_C - 8\cdot y_C + 43 = 0 \end{vmatrix}$ (-) $\Rightarrow 13\cdot y_C - 52 = 0$

$y_C = 4$

$x_C = -11$

$C(-11, 4)$

б) $A(4,1)$, $B(3,-4)$, $C(-11,4)$ $\quad S_{\triangle ABC} = ?$

$S_{\triangle ABC} = \frac{1}{2}\cdot\left|\begin{matrix} 4 & 1 & 1 \\ 3 & -4 & 1 \\ -11 & 4 & 1 \end{matrix}\right| = \frac{1}{2}\cdot|-16+12-11-$ (31, 47)

$-44-3-16| =$

ОКС Oxy

$= \frac{1}{2}\cdot|-15-47-16| = \frac{1}{2}\cdot|78| = 39$ кв. ед.

в) Описана около $\triangle ABC$ окр.

? координ. на центъра S и $R = ?$ на описаната окръжност

I Н. т. $S = S_{AB} \cap S_{BC} \cap S_{AC}$

II Н. Нека т. $S(x,y) \Rightarrow |\vec{SA}| = |\vec{SB}| = |\vec{SC}|$

$\vec{SA}(x-4, y-1)$

1) $S_{AC} = ?$ $\quad AC: x+5y-9=0$, N - средата на AC $\quad A(4,1)$, $C(-11,4) \Rightarrow N\left(-\frac{7}{2}, \frac{5}{2}\right)$

$S_{AC}: 5\cdot x - y + D = 0$

$N\left(-\frac{7}{2}, \frac{5}{2}\right) \Rightarrow 5\cdot\left(-\frac{7}{2}\right) - \frac{5}{2} + D = 0 \quad D = 20$

$S_{AC}: 5x - y + 20 = 0$

2) $S_{BC} = ?$ $\quad B(3,-4)$, $C(-11,4) \Rightarrow M(-4,0)$ - средата на BC

$BC: \left|\begin{matrix} x & y & 1 \\ 3 & -4 & 1 \\ -4 & 0 & 1 \end{matrix}\right| = 0 \Rightarrow -4x - 4y - 16 - 3y = 0 \quad -4x - 7y - 16 = 0 \,/\cdot(-1)$

$4x + 7y + 16 = 0$ да

3 $\quad$ -4

-4 $\quad$ 0

S_{BC} $\begin{cases} \ni M(-4,0) \\ \perp BC: 4x+7y+16=0 \end{cases} \Rightarrow S_{BC}: 7x - 4y + D = 0$

$7\cdot(-4) + 0 + D = 0 \quad D = 28$

1S $\begin{vmatrix} S_{BC}: 7x - 4y + 28 = 0 \\ S_{AC}: 5x - y + 20 = 0 \end{vmatrix} \Rightarrow$ т. $S(-4,0) \equiv M(-4,0)$

$R = |\vec{SA}| = \sqrt{64+1} = \sqrt{65}$

$A(4,1)$

$\vec{SA}(8,1)$

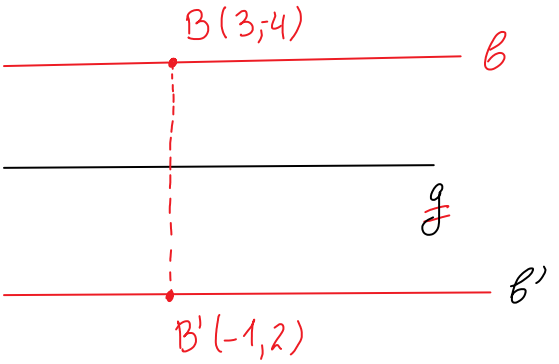
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9 зад. ОКС Oxy

$g: 2x-3y-5=0$

Да се намери образът на права b под действие на σ_g.

$b \xrightarrow{\sigma_g} b'$, търсим b'.



а) $b: 2x-3y-18=0 \Rightarrow b \parallel g$

$\Rightarrow b' = \sigma_g(b)$, $b' \parallel b \parallel g$

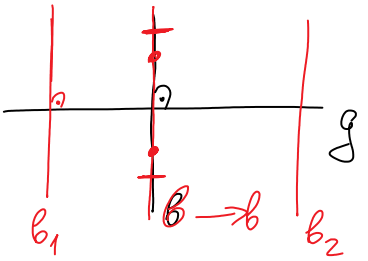
Изб. т. $B \in b$, $B(3,-4) \xrightarrow{\sigma_g} B'(-1,2)$

$b' \begin{cases} \parallel b \\ \ni B'(-1,2) \end{cases} \Rightarrow$ Упр.

б) $g: 2x-3y-5=0$

$b: 3x+2y-1=0 \Rightarrow g \perp b$

$b \xrightarrow{\sigma_g} b$



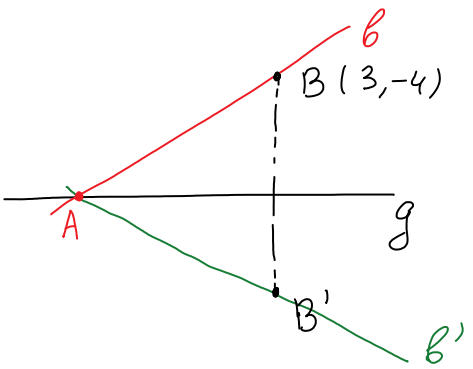
в) $g: 2x-3y-5=0$

$(b: 5x-y-19=0)$ $b \# g$, $b \not\perp g$

$A = g \cap b \Rightarrow A(4,1)$

Изб. т. $B \in b$, изб. $B(3,-4) \xrightarrow{\sigma_g} B'(-1,2)$

$b' \begin{cases} \ni A(4,1) \\ \ni B'(-1,2) \end{cases} \Rightarrow b': x+5y-9=0$ Да



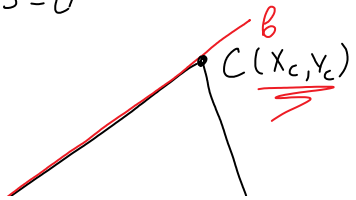
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10 зад. ОКС Oxy

$b: 2x-y=0$

$c: x-2y+3=0$

т. $M(3,4)$

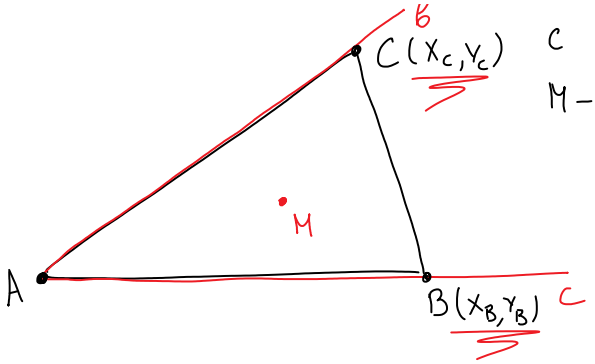


а) ?, коорд. на върховете на $\triangle ABC$:

b съдържа AC

c съдържа AB

M – медицентърът на $\triangle ABC$



С средища AB

M – медицентърът на $\triangle ABC$

1) $A = B \cap C \Rightarrow A(1,2)$ (Y пр.)

2) $\frac{X_A + X_B + X_C}{3} = X_M$, $\frac{Y_A + Y_B + Y_C}{3} = Y_M$ $M(3,4)$

$\frac{1 + X_B + X_C}{3} = 3$ $\frac{2 + Y_B + Y_C}{3} = 4$

$X_B + X_C = 8$ $Y_B + Y_C = 10$

3) $B \in c \Rightarrow X_B - 2Y_B + 3 = 0$

$C \in b \Rightarrow 2X_C - Y_C = 0$

$\Rightarrow$

$X_B + X_C = 8$

$Y_B + Y_C = 10$

$X_B - 2Y_B + 3 = 0$

$2X_C - Y_C = 0$

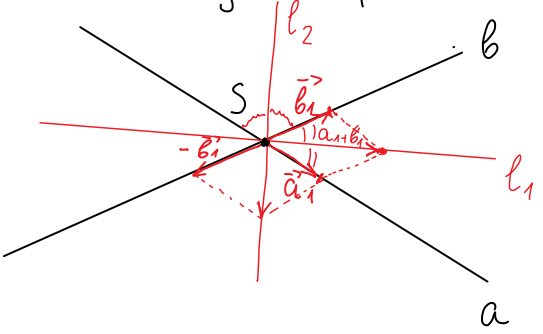
Отг. $B(5,4)$

$C(3,6)$

б) !!! Да се намерят.

* $S_{\triangle ABC} = ?$
* $P_{\triangle ABC} = ?$
* вида на $\triangle ABC$ спр. ъглите му
* координати на ортоцентъра
* коорд. на ц-р на описаната около $\triangle ABC$ окр.

Уравнения на ъглополовящи на ъгли между 2 прави (ъглоп. на тъп и остър ъгъл)



1) т. $S = a \cap b$

2) $a \parallel \vec{a}$, $|\vec{a}| \Rightarrow \vec{a_1} = \frac{\vec{a}}{|\vec{a}|} \Rightarrow |\vec{a_1}| = 1$

$b \parallel \vec{b}$, $|\vec{b}| \Rightarrow \vec{b_1} = \frac{\vec{b}}{|\vec{b}|} \Rightarrow |\vec{b_1}| = 1$

$l_1 \begin{cases} \parallel \vec{a_1} + \vec{b_1} \\ \ni S \end{cases}$, $l_2 : \begin{cases} \parallel \vec{a_1} - \vec{b_1} \\ \ni S \end{cases}$

3) Тъп и остър ъгъл

$l_1 \to \sphericalangle(\vec{a_1}, \vec{b_1}) \Rightarrow (\vec{a_1}.\vec{b_1}) = ? = \cos\sphericalangle(\vec{a_1}, \vec{b_1})$

1сл. Ако $(\vec{a_1}.\vec{b_1}) > 0 \Rightarrow$ l_1 е ъглополовяща на остър $\sphericalangle(a,b)$

l_2 е –"– на тъпия $\sphericalangle(a,b)$

2сл. Ако $(\vec{a_1}.\vec{b_1}) < 0 \Rightarrow$ $l_1 \to$ тъп $\sphericalangle(a,b)$

$l_2 \to$ остър $\sphericalangle(a,b)$

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11 зад. ОКС Oxy

$a: 3x-4y+5=0$

$b: 4x-3y-5=0$ $\quad a \not\parallel b$

? уравн. на l_1 и l_2 – ъглополов. на $\sphericalangle(a,b)$

Да се определи кой е ъглоп. на острия $\sphericalangle$, кой е ъглоп. на тъпия $\sphericalangle(a,b)$

1) $S = a \cap b \Rightarrow \begin{vmatrix} 3x-4y+5=0 \\ 4x-3y-5=0 \end{vmatrix} \Rightarrow S(5,5)$

2) $a: 3x-4y+5=0 \Rightarrow a \parallel \vec{a}(-B,A) \Rightarrow \vec{a}(4,3) \Rightarrow |\vec{a}| = \sqrt{4^2+3^2} = 5$

$\Rightarrow \vec{a_1}\left(\frac{4}{5}, \frac{3}{5}\right)$, $|\vec{a_1}| = 1$ ✓

$\cos\sphericalangle(\vec{a_1}, \vec{e_1})$ $\quad$ $\cos\sphericalangle(\vec{a_1}, \vec{e_2})$

$b: 4x-3y-5=0 \Rightarrow b \parallel \vec{b}(3,4) \Rightarrow |\vec{b}| = 5 \Rightarrow \vec{b_1}\left(\frac{3}{5}, \frac{4}{5}\right)$ ✓

3) $l_1 \begin{cases} \ni S(5,5) \\ \parallel \vec{a_1} + \vec{b_1}\left(\frac{7}{5}, \frac{7}{5}\right) \end{cases} \Rightarrow$ $(1,1) \parallel (\vec{a_1}+\vec{b_1})$

$l_1: \begin{cases} x = 5 + s.1 \\ y = 5 + s.1 \end{cases} \begin{matrix} s \in \mathbb{R} \\ - \end{matrix} \Rightarrow l_1: x-y=0$

$l_2 \begin{cases} \ni S(5,5) \\ \parallel (\vec{a_1}-\vec{b_1})\left(\frac{1}{5}, -\frac{1}{5}\right) \end{cases} \Rightarrow$ $l_2 \begin{cases} x = 5 + p.1 \\ y = 5 + p.(-1) \end{cases}$ $(+) \Rightarrow l_2: x+y-10=0$

4) $l_1 \to \sphericalangle(\vec{a_1}, \vec{b_1})$ $\quad (\vec{a_1}.\vec{b_1}) = \frac{4}{5}\cdot\frac{3}{5} + \frac{3}{5}\cdot\frac{4}{5} = \frac{24}{25} > 0 \Rightarrow$

$\sphericalangle(\vec{a_1}, \vec{b_1})$ – остър $\Rightarrow$ l_1 е ъглоп. на острия $\sphericalangle(a,b)$

l_2 – " – –"– тъпия $\sphericalangle(a,b)$

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12 зад. ОКС Oxy

а) $A(1,2)$, $B(-1,3)$, $C(5,4)$

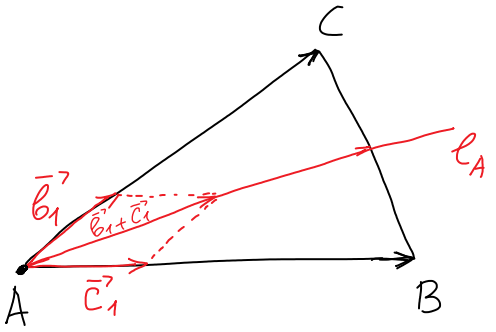
?, уравнение на ℓ_A - ъглоп. на вътр. ъгъл при т. A на $\triangle ABC$

$\ell_A \ni A(1,2)$

$\ell_A \to \measuredangle(\vec{AB}, \vec{AC})$

$\vec{AB}(-2,1) \Rightarrow |\vec{AB}| = \sqrt{(-2)^2+1^2} = \sqrt{5}$

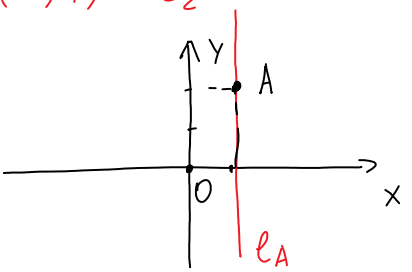
$\vec{c_1} = \frac{\vec{AB}}{|\vec{AB}|} \Rightarrow \vec{c_1}\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$



$\vec{AC}(4,2) \Rightarrow |\vec{AC}| = \sqrt{20} = 2\sqrt{5} \Rightarrow \vec{b_1}\left(\frac{4}{2\sqrt{5}}, \frac{2}{2\sqrt{5}}\right) \Rightarrow \vec{b_1}\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

$\left(\vec{b_1}+\vec{c_1}\right)\left(0, \frac{2}{\sqrt{5}}\right) \parallel (0,1) \to \vec{e_2}$

$\ell_A \begin{cases} \ni A(1,2) \\ \parallel \vec{e_2}(0,1) \end{cases}$



$\ell_A : x = 1$

б) (Упр.) $A(1,-2)$, $B(2,0)$, $C\left(-\frac{2}{3}, \frac{4}{3}\right)$

? коорд. на центъра I на вписаната в $\triangle ABC$ окръжност и r=?

Отг. т. $I\left(1, -\frac{1}{3}\right)$, $r = \frac{\sqrt{5}}{3}$